📋 DAILY WORK REPORT — DATACENTER INFRASTRUCTURE

Proxmox VE HA Cluster & Ceph/ZFS Storage Validation

Report Field	Details
Date of Execution	September 24, 2026 (24-09-2026)
Engineer Name	Nishu Saini / CISAI Infrastructure & Cloud Engineering
Location	Production Datacenter / On-Premises Server Hall
Target Infrastructure	3-Node Proxmox VE 8.x Enterprise HA Cluster (`cisai-pve`)
Target Nodes	`prox22` (`10.31.6.172`), `prox23` (`10.31.6.173`), `prox24` (`10.31.6.174`)
Storage Architecture	12x 20TB Enterprise SAS Ceph Reef + 3x 127 TiB ZFS RAIDZ2 Pools
Overall Status	**100% COMPLETED / PRODUCTION SIGN-OFF (HEALTH_OK)**

1. Executive Summary

On **September 24, 2026**, full-day on-site physical datacenter validation, fault-tolerance stress testing, and disaster recovery verification were conducted on the 3-node enterprise Proxmox VE cluster (`cisai-pve`).

All planned testing scenarios from the master datacenter runbook (TC-01 through TC-19, storage network blackouts, dual-drive physical pulls, and live VM stress migrations) were successfully executed with **zero data corruption, zero permanent service outage, and 100% cryptographic data integrity**.

2. Key Accomplishments & Tasks Executed Today

A. Direct Field Access & Network Fabric Verification

Configured dedicated direct laptop connection port on `prox22` (`nic0` : `192.168.99.1/24`) to guarantee non-disruptive, high-speed on-site management at `https://192.168.99.1:8006` .

Verified full-mesh 10G SFP+ DAC storage interconnect (`192.168.10.x`) with ultra-low latency (**0.04ms – 0.14ms**) and 0% packet drop.

Confirmed Out-of-Band IPMI/BMC reachability on `prox22` (`10.31.5.78`), `prox23` (`10.31.5.61`), and `prox24` (`10.31.5.62`).

B. Master 19-Scenario Physical Testing Suite (TC-01 – TC-19)

Ceph Storage Fault Injection (TC-01 – TC-07):

* Single SAS disk pull, 2-disk simultaneous pull, 3-disk pull, and drive slot mobility swap (Slot 1 ↔ Slot 3).

* Ceph CRUSH rules and $3\times$ replication ($min_size=2$) maintained continuous direct I/O without virtual machine lockup.

ZFS Storage Fault Injection (TC-12 – TC-14):

* Tested mirrored ZFS OS SSD drive pull on `prox23` and software disk failure injection on `prox24` . Automatic resilver completed with 0 read/write/cksum errors.

Network & Power Redundancy (TEST-5.1, TC-08, TC-10, TC-11):

* Unplugged single PSU power feeds to verify dual-feed A/B power transfer.

* Disconnected 10G DAC cables and 1G management links to validate Linux hardware watchdog self-fencing and Corosync quorum split-brain protection (TC-09).

C. Live VM High-Concurrency FIO Workload & Stress Migration

Injected synthetic 4K direct async I/O stress (**1.2+ Million IOPS peak baseline**) on VM 9003.

Executed online live migration across nodes (`prox22` ➡️ `prox23`) under 100% write duty cycle.

Live migration finished in **1.08 seconds** with **< 120ms I/O freeze** and **100% SHA256 cryptographic verification (0 corrupted blocks)**.

D. ZFS Storage Scrub & Bitrot Self-Healing Audit

Conducted a full cryptographic tree scrub across the 127 TiB ZFS RAIDZ2 storage pool (`zfs-ha`).

Verified **0 READ errors, 0 WRITE errors, and 0 CKSUM (bitrot) errors** across all 21 SAS hard drives.

E. Root Cause Analysis (RCA) & Operational SOP Documentation

Identified and documented the Linux kernel dynamic SCSI re-enumeration (`sdb` ➡️ `sdn`) and LVM device-mapper linear mapping desync during physical drive hot-swaps.

Formulated the standard automated SOP to instantly refresh device-mapper tables without requiring server reboots.

Resolved Web UI transition artifacts (blue wrench & question mark icons) associated with HA maintenance mode and `pvestatd`.

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3. Final Cluster Health Sign-Off Matrix

Subsystem	Target State	Final Measured Status on 24-09-2026	Sign-Off
Node `prox22`	Online / CRM Master	🟢 **ONLINE** (192 Cores, 1 TiB RAM, Uptime Clean)	**VERIFIED**
Node `prox23`	Online / Member	🟢 **ONLINE** (192 Cores, 1 TiB RAM, Uptime Clean)	**VERIFIED**
Node `prox24`	Online / Member	🟢 **ONLINE** (192 Cores, 1 TiB RAM, Uptime Clean)	**VERIFIED**
Corosync Cluster	Quorum 3/3 Votes	🟢 **3 / 3 Votes (`Quorate: YES` , `Ring: 1.10f`)**	**VERIFIED**
Ceph Storage	`HEALTH_OK`	🟢 **12 / 12 OSDs UP & IN, 33/33 PGs active+clean**	**VERIFIED**
ZFS Storage Pools	`zfs-ha` RAIDZ2 (127 TiB)	🟢 **100% ONLINE (READ: 0, WRITE: 0, CKSUM: 0)**	**VERIFIED**
Proxmox HA Manager	CRM / LRM Armed	🟢 **Quorum OK / Watchdogs Armed & Active**	**VERIFIED**
Virtual Machines	Workloads Active	🟢 `VM 9002` (Running), `VM 9003` (Running)	**VERIFIED**
Data Integrity	Zero Corruption	🟢 **0 Bytes Lost / 100% Verified**	**VERIFIED**

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4. Deliverables & Documentation Index

All test scripts, executive reports, and audit PDFs are saved in `/home/nishusaini/Desktop/Proxmox_Testing_Reports/` :

1. 📕 **Daily Work Report (24-09-2026):** `DAILY_WORK_REPORT_2026-09-24.pdf`
2. 📕 **Root Cause Analysis (RCA) Report:** `CEPH_STORAGE_TESTING_RCA_REPORT.pdf`
3. 📕 **Live VM FIO Workload & Stress Failover Report:** `vm_fio_live_failover_report_20260924_223442.pdf`
4. 📕 **ZFS Storage Scrub & Bitrot Audit Report:** `zfs_scrub_bitrot_report_20260924_224147.pdf`
5. 📕 **Simultaneous 10G Outage Blackout Report:** `simultaneous_10g_blackout_report_20260924_221330.pdf`
6. 📕 **Full 19-Scenario Master Checklist Report:** `proxmox_testing_report_20260924_214255.pdf`

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5. Engineer Sign-Off

Work Completed By: Nishu Saini

Date & Timestamp: 2026-09-24 22:48:00 IST

Final Verdict: DATACENTER TESTING 100% SUCCESSFUL — CLUSTER IS IN PRODUCTION READY STATE